

Day 15 – 3D Avatar Generation & Human Modeling Research

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Objective

The objective of today's work was to explore 3D avatar generation and human modeling techniques used in modern Virtual Try-On systems. The focus was on understanding how AI-based methods reconstruct human body shapes, generate realistic avatars, and support future virtual clothing applications.

Introduction

Virtual Try-On systems are increasingly adopting 3D technologies to provide more realistic clothing visualization. Unlike traditional 2D systems, 3D avatar generation enables clothing to be displayed on a digital representation of the user. This improves personalization, fitting accuracy, and user experience.

The purpose of this study was to investigate major technologies used for human body reconstruction and avatar generation, understand their workflows, and analyze their suitability for future Virtual Try-On development.

3D Avatar Generation Overview

3D avatar generation is the process of creating a digital human representation from images, body measurements, or other user inputs.

A typical avatar generation pipeline includes:

- User image acquisition
- Body detection

- Feature extraction
- Human body reconstruction
- Mesh generation
- Avatar creation
- Virtual Try-On integration

These avatars can then be used for clothing visualization, AR applications, gaming, and e-commerce solutions.

SMPL (Skinned Multi-Person Linear Model)

SMPL is a widely used parametric human body model that represents the human body using shape and pose parameters.

Working Principle

- Receives body shape and pose information.
- Generates a realistic 3D human mesh.
- Supports animation and body movement.

Advantages

- Widely used in research.
- Compact representation.
- Supports different body shapes.
- Easy integration with computer vision systems.

Limitations

- Limited clothing details.
- Requires additional processing for realistic garments.
- Does not generate detailed facial features.

Applications

- Human body modeling
- Pose estimation research
- Avatar generation

- Virtual Try-On systems

PIFuHD (Pixel-Aligned Implicit Function for High Resolution Human Digitization)

PIFuHD is a deep learning framework that reconstructs detailed 3D human models from a single RGB image.

Working Principle

- Uses a single image as input.
- Extracts body and clothing information.
- Reconstructs a detailed 3D human model.

Advantages

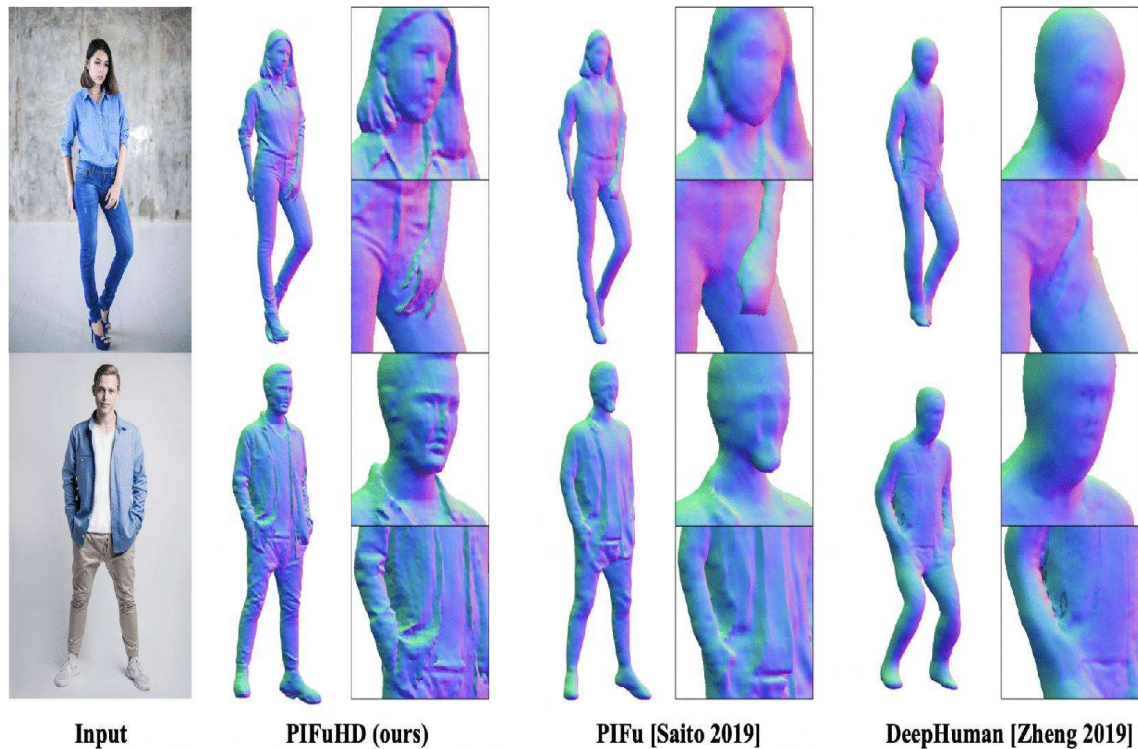
- High-quality reconstruction.
- Preserves clothing details.
- Generates realistic body shapes.

Limitations

- Computationally expensive.
- Slower processing time.
- Requires GPU resources.

Applications

- High-quality avatar generation
- Digital human creation
- Virtual fashion systems



Human Mesh Recovery (HMR)

Human Mesh Recovery is a technique that predicts a 3D human mesh directly from a 2D image.

Working Principle

- Takes RGB images as input.
- Estimates body pose and shape.
- Generates a complete 3D human mesh.

Advantages

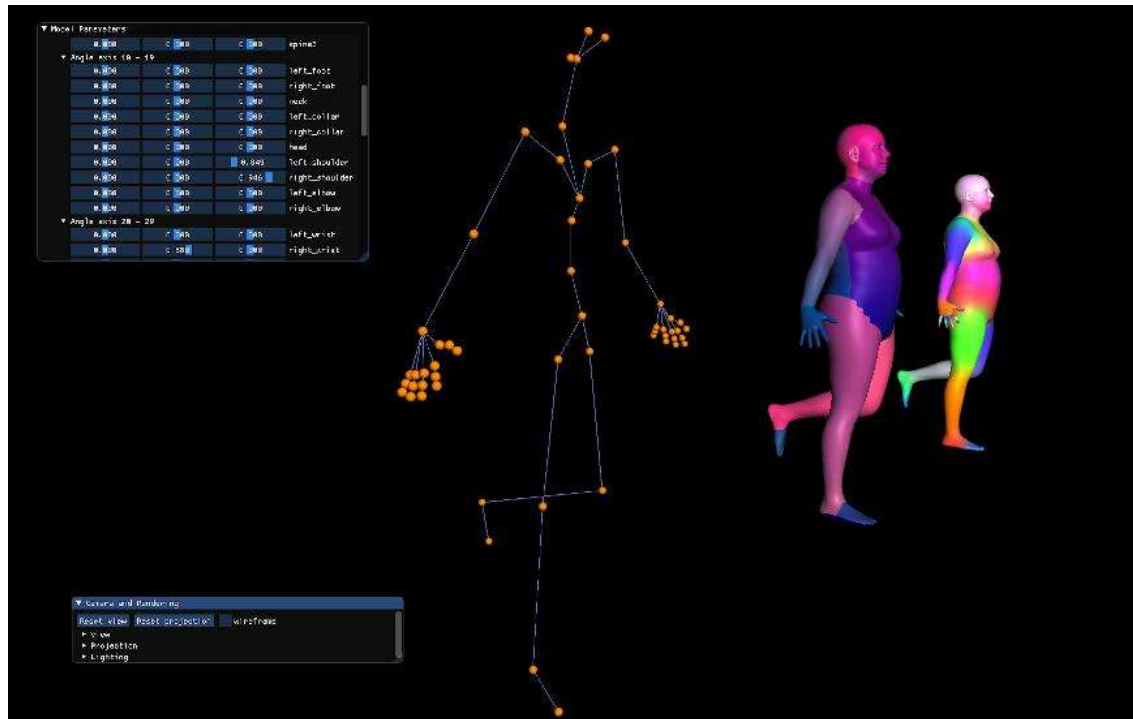
- Fast processing.
- Good balance between speed and accuracy.
- Suitable for large-scale applications.

Limitations

- Lower detail compared to PIFuHD.
- Accuracy depends on image quality.

Applications

- Human pose analysis
- Avatar generation
- Human reconstruction



Ready Player Me

Ready Player Me is an avatar creation platform that generates customizable avatars for virtual worlds and AR/VR applications.

Working Principle

- User uploads a photo.
- System generates a digital avatar.
- Avatar can be customized and exported.

Advantages

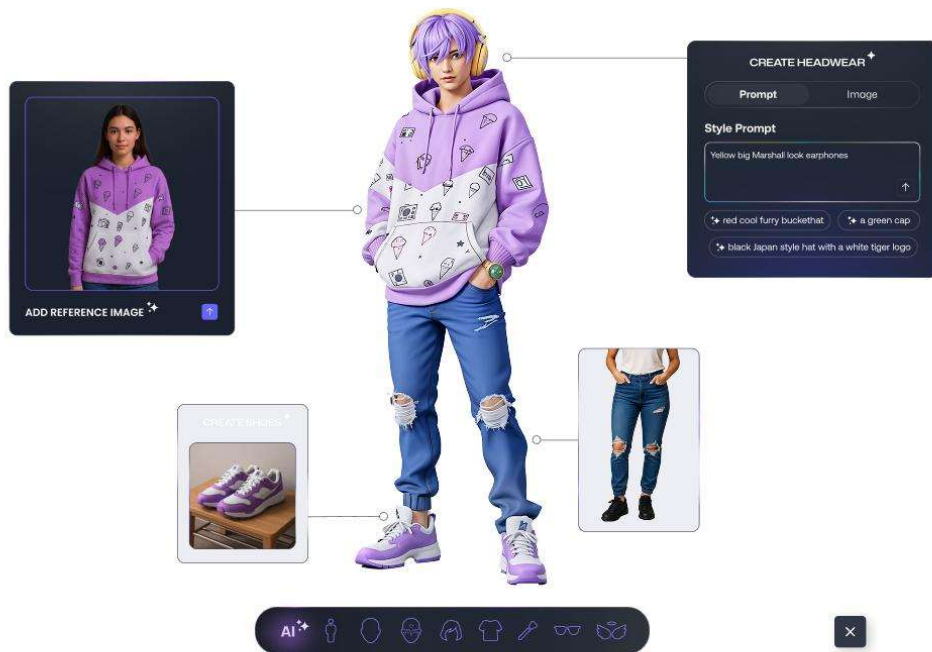
- Easy to use.
- Fast avatar generation.
- Suitable for beginners.
- Compatible with many platforms.

Limitations

- Lower realism compared to research models.
- Limited body reconstruction accuracy.

Applications

- Virtual worlds
- Metaverse applications
- AR/VR experiences
- Gaming environments



Technology Comparison

Technology	Input	Output	Accuracy	Processing Time
SMPL	Body Parameters	3D Human Mesh	High	Fast
PIFuHD	Single RGB Image	Detailed Human Model	Very High	Slow
HMR	RGB Image	Human Mesh	High	Medium
Ready Player Me	User Photo	Avatar	Medium	Fast

3D Avatar Generation Workflow

User Image



Body Detection



Feature Extraction



3D Reconstruction



Human Mesh Generation



Avatar Creation



Virtual Try-On System

Workflow Explanation

1. User uploads an image.
2. Human body is detected.
3. Important body features are extracted.
4. AI reconstructs the body structure.
5. A 3D mesh is generated.
6. The mesh is converted into an avatar.
7. The avatar is used for virtual clothing visualization.

Blender Exploration and Avatar Visualization

To understand avatar-based systems, Blender was explored as a 3D visualization tool.

Activities performed:

- Opened Blender workspace.
- Imported FBX avatar models.
- Visualized avatars in 3D space.
- Explored Material Preview mode.
- Rotated and inspected avatar models.
- Studied avatar structure and hierarchy.

The exploration provided practical exposure to handling avatar assets used in Virtual Try-On systems.

Example Outputs

Figure 1

Blender Workspace

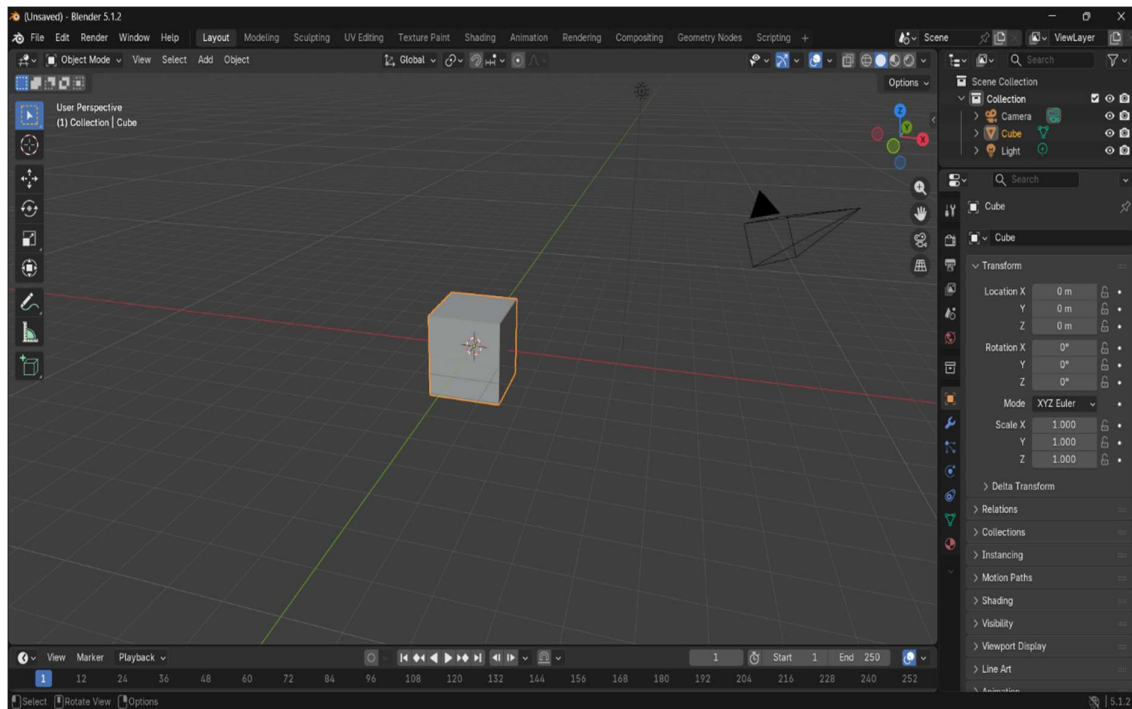


Figure 2

Imported Human Avatar Model



Figure 3

Avatar in Material Preview Mode

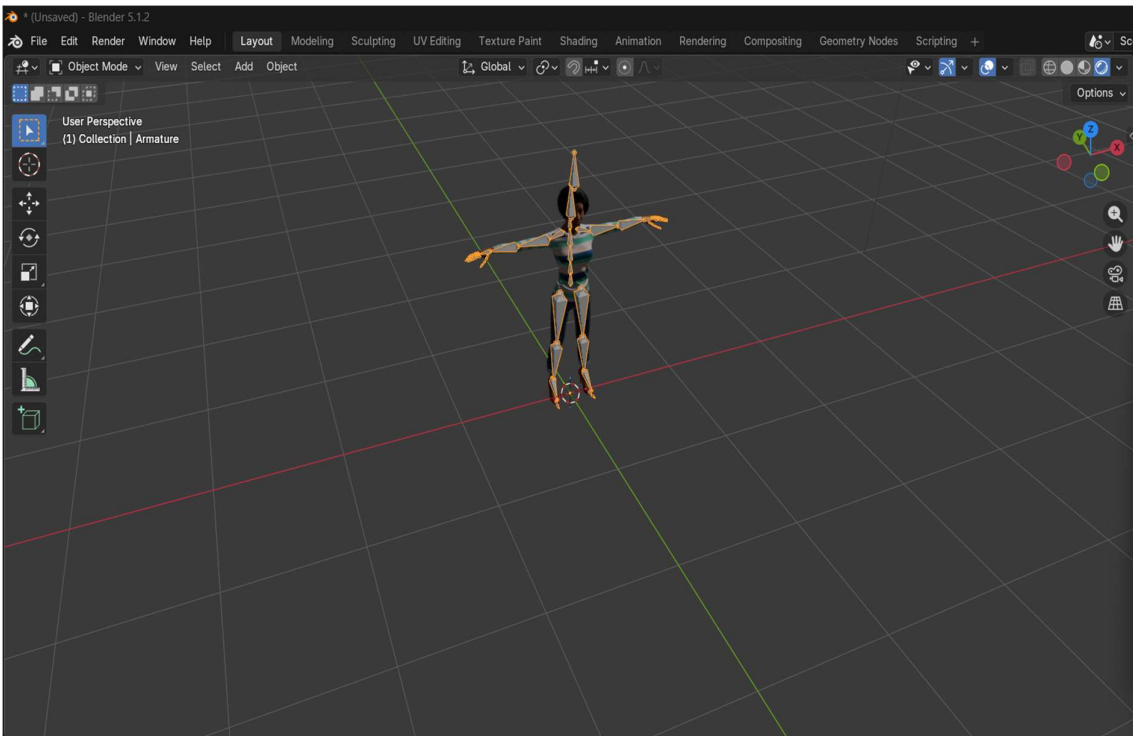
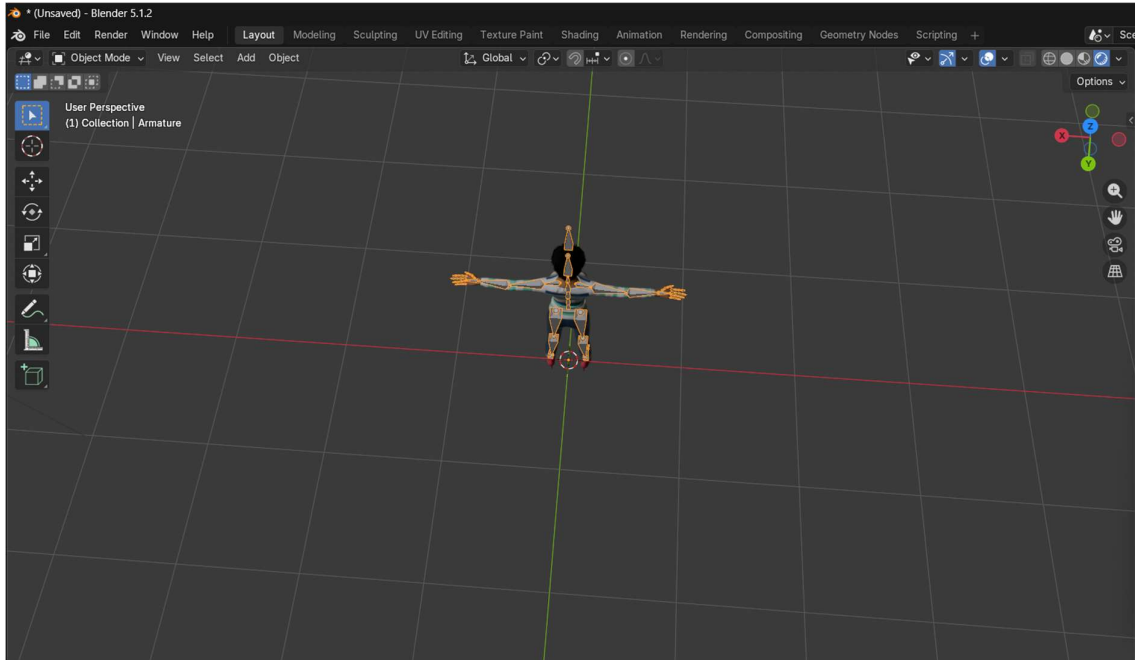


Figure 4

Avatar Visualization from Different Perspectives

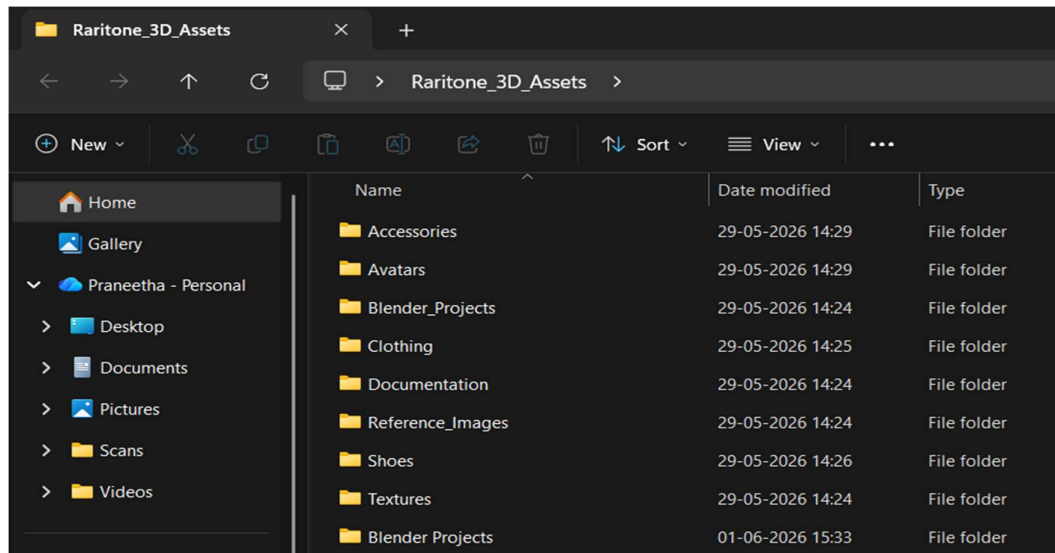


Dataset Collection Update

The existing 3D asset repository was reviewed and updated.

Current categories include:

- Clothing
- Shoes
- Accessories
- Male Avatars
- Female Avatars
- Reference Images
- Blender Projects
- Textures
- Documentation



These assets will support future research and development activities related to Virtual Try-On systems and avatar generation.

Recommendation

Based on the research conducted:

- Ready Player Me is suitable for rapid avatar generation and prototyping.
- SMPL is suitable for body modeling and academic research.
- HMR provides a balance between speed and reconstruction quality.
- PIFuHD produces the highest visual quality but requires significant computational resources.

For future Virtual Try-On development, a combination of SMPL-based body modeling and advanced reconstruction techniques such as PIFuHD can provide realistic and accurate avatar generation.

Conclusion

This study explored major technologies used for 3D avatar generation and human modeling. Research was conducted on SMPL, PIFuHD, Human Mesh Recovery, and Ready Player Me. A workflow for avatar generation was analyzed, and Blender was used to visualize avatar assets. The findings provide a strong foundation for future work involving 3D Virtual Try-On systems, avatar customization, and AR/VR-based fashion applications.